ALL Yaml files explanation

1. deployment.yaml >> Application deployment kosam

```
apiVersion: apps/v1
kind: Deployment
metadata:
 name: netflix
spec:
 replicas: 3
 selector:
  matchLabels:
   app: netflix
 template:
  metadata:
   labels:
    app: netflix
  spec:
   containers:
    - name: netflix
      image: mycompany/netflix:v1
      ports:
       - containerPort: 80
```

explanation:

apiVersion: apps/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: Deployment > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada Deployment object create chestunnam.

metadata: > information/details
→ Object ki sambandhinchina basic information ikkada untundi.

name: netflix > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix → Deployment peru netflix.

spec: > kavalsina setup
→ Manaki application ela undalo, em configuration kavalo ikkada cheptham.

replicas: 3 > enni Pods?
→ Enni Pods run avvali ani chepthundi.
→ replicas: 3 → 3 Pods run avvali.

selector: > ye Pods ni identify cheyyali?
→ Ee Deployment ye Pods ni manage cheyyalo identify chestundi.

matchLabels: > ye label match avvali?
→ Pod lo unna label tho match chesi correct Pods ni identify chestundi.

app: netflix > Pod label
→ Pod ki app=netflix ane label untundi.

template: > Pod blueprint
→ Kubernetes create chese Pod ela undalo ikkada cheptham.

metadata: > Pod information/details
→ Create ayye Pod ki sambandhinchina information ikkada untundi.

labels: > Pod labels/tags
→ Pod ni identify cheyyadaniki labels ikkada define chestham.

app: netflix > Pod label
→ Create ayye prathi Pod ki app=netflix label vastundi.

spec: > Pod kavalsina setup
→ Pod lopala em run avvali ane configuration ikkada untundi.

containers: > Pod lo em run avvali?
→ Pod lopala run ayye container details ikkada define chestham.

name: netflix > container peru
→ Container ki netflix ane peru pettam.

image: mycompany/netflix:v1 > ye Docker image run cheyyali?
→ Container lopala mycompany/netflix:v1 Docker image run avtundi.
→ v1 → image version/tag.

ports: > container ports
→ Container lo application use chese ports ikkada define chestham.

containerPort: 80 > application/container port
→ Container lo application port 80 lo run/listen chestundi ani chepthundi.

---

2. service.yaml >> Pod networking kosam

```
apiVersion: v1
kind: Service
metadata:
  name: netflix-service
spec:
  selector:
    app: netflix
  ports:
    - port: 80
      targetPort: 80
  type: ClusterIP
```

explanation:
apiVersion: v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: Service > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada Service object create chestunnam.

metadata: > information/details
→ Service ki sambandhinchina basic information ikkada untundi.

name: netflix-service > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix-service → Service peru netflix-service.

spec: > kavalsina setup
→ Service ela work avvali ane configuration ikkada cheptham.

selector: > ye Pods ni identify cheyyali?
→ Ee Service ye Pods ki traffic pampalo identify chestundi.

app: netflix > Pod label
→ app=netflix label unna Pods ni Service select chestundi.

ports: > ports configuration
→ Service use chese ports ikkada define chestham.

port: 80 > Service port
→ Service port 80 lo traffic receive chestundi.

targetPort: 80 > Pod/container port
→ Service vachina traffic ni selected Pod lo port 80 ki pampistundi.

type: ClusterIP > Service type
→ Service ni cluster lopala access cheyyadaniki ClusterIP type use chestunnam.

---

3. ingress.yaml >> External traffic routing kosam

```
apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: netflix-ingress
spec:
  rules:
    - host: netflix.example.com
      http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: netflix-service
                port:
                  number: 80
```

explanation:
apiVersion: networking.k8s.io/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: Ingress > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada Ingress object create chestunnam.

metadata: > information/details
→ Ingress ki sambandhinchina basic information ikkada untundi.

name: netflix-ingress > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix-ingress → Ingress peru netflix-ingress.

spec: > kavalsina setup
→ External request ela route avvali ane configuration ikkada cheptham.

rules: > routing rules
→ Incoming request ekkadiki vellalo rules ikkada define chestham.

host: netflix.example.com > domain
→ Ee domain ki request vachinappudu ee rule apply avtundi.

http: > HTTP configuration
→ HTTP request ki sambandhinchina configuration ikkada untundi.

paths: > URL paths
→ Ye URL path request ni handle cheyyalo ikkada define chestham.

path: / > URL path
→ Root path / ki vachina request ni handle chestundi.

pathType: Prefix > path matching
→ / tho start ayye paths ni match chestundi.

backend: > request ekkadiki vellali?
→ Match ayina request ni ye Service ki pampalo chepthundi.

service: > Service details
→ Request ni Kubernetes Service ki pampistunnam.

name: netflix-service > Service peru
→ Request netflix-service ane Service ki vellali.

port: > Service port
→ Request Service lo ye port ki vellalo chepthundi.

number: 80 > port number
→ Request Service port 80 ki vellali.

---

4. configmap.yaml >> Normal configuration kosam

```
apiVersion: v1
kind: ConfigMap
metadata:
  name: netflix-config
data:
  APP_ENV: production
  API_URL: https://api.netflix.example.com
  LOG_LEVEL: info
```

explanation:
apiVersion: v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: ConfigMap > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada ConfigMap object create chestunnam.

metadata: > information/details
→ ConfigMap ki sambandhinchina basic information ikkada untundi.

name: netflix-config > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix-config → ConfigMap peru netflix-config.

data: > configuration values
→ Application ki kavalsina normal configuration values ikkada store chestham.

APP_ENV: production > application environment
→ Application production environment lo run avtundi ani chepthundi.

API_URL: https://api.netflix.example.com > API URL
→ Application use cheyyalsina API URL ni chepthundi.

LOG_LEVEL: info > logging level
→ Application logs ye level lo undalo chepthundi.
→ info → information level logs.

---

5. secret.yaml >> Sensitive data kosam

```
apiVersion: v1
kind: Secret
metadata:
  name: netflix-secret
type: Opaque
stringData:
  DB_USERNAME: netflix
  DB_PASSWORD: example-password
```

explanation:
apiVersion: v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: Secret > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada Secret object create chestunnam.

metadata: > information/details
→ Secret ki sambandhinchina basic information ikkada untundi.

name: netflix-secret > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix-secret → Secret peru netflix-secret.

type: Opaque > Secret type
→ General sensitive data kosam Opaque type use chestunnam.

stringData: > secret values
→ Secret values ni readable string format lo ikkada provide chestham.

DB_USERNAME: netflix > database username
→ Database ki connect avvadaniki username netflix ani chepthundi.

DB_PASSWORD: example-password > database password
→ Database ki connect avvadaniki password example-password ani chepthundi.

---

6. hpa.yaml >> Automatic Pod scaling kosam

```
apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
 name: netflix-hpa
spec:
 scaleTargetRef:
   apiVersion: apps/v1
   kind: Deployment
   name: netflix
 minReplicas: 2
 maxReplicas: 10
 metrics:
   - type: Resource
     resource:
       name: cpu
       target:
         type: Utilization
         averageUtilization: 70
```

explanation:
apiVersion: autoscaling/v2 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: HorizontalPodAutoscaler > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada HPA object create chestunnam.

metadata: > information/details
→ HPA ki sambandhinchina basic information ikkada untundi.

name: netflix-hpa > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix-hpa → HPA peru netflix-hpa.

spec: > kavalsina setup
→ Pods ni ela automatically scale cheyyalo ikkada cheptham.

scaleTargetRef: > ye object ni scale cheyyali?
→ HPA ye object ni scale cheyyalo chepthundi.

apiVersion: apps/v1 > target version/rules
→ Target object ye API version/rules use chestundo chepthundi.

kind: Deployment > target object
→ Scale cheyyalsina object Deployment ani chepthundi.

name: netflix > target object peru
→ netflix ane Deployment ni scale cheyyali.

minReplicas: 2 > minimum Pods
→ Minimum ga 2 Pods undali.

maxReplicas: 10 > maximum Pods
→ Maximum ga 10 Pods varaku increase avvachu.

metrics: > scaling metrics
→ Pods ni ye measurement batti scale cheyyalo chepthundi.

type: Resource > resource metric
→ Kubernetes resource usage ni scaling kosam use chestunnam.

resource: > resource details
→ Ye resource ni monitor cheyyalo chepthundi.

name: cpu > CPU resource
→ CPU usage ni monitor chestunnam.

target: > target value
→ CPU ki target value ikkada define chestham.

type: Utilization > usage percentage
→ CPU usage percentage ni target ga use chestunnam.

averageUtilization: 70 > CPU target
→ Average CPU usage 70% target ga pettam.

---

7. pvc.yaml >> Persistent storage kosam

```
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: netflix-storage
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 10Gi
```

explanation:
apiVersion: v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: PersistentVolumeClaim > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada PVC object create chestunnam.

metadata: > information/details
→ PVC ki sambandhinchina basic information ikkada untundi.

name: netflix-storage > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix-storage → PVC peru netflix-storage.

spec: > kavalsina setup
→ Manaki kavalsina storage configuration ikkada cheptham.

accessModes: > storage access mode
→ Storage ni ela access cheyyalo chepthundi.

ReadWriteOnce > access mode
→ Oka node nunchi volume ni read/write cheyyadaniki request chestunnam.

resources: > storage resources
→ Manaki kavalsina storage resources ikkada define chestham.

requests: > storage request
→ Entha storage kavalo request chestunnam.

storage: 10Gi > storage size
→ Manaki 10Gi storage kavali ani request chestunnam.

---

8. namespace.yaml >> Resources separate cheyyadaniki

```
apiVersion: v1
kind: Namespace
metadata:
  name: netflix-prod
```

explanation:
apiVersion: v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: Namespace > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada Namespace object create chestunnam.

metadata: > information/details
→ Namespace ki sambandhinchina basic information ikkada untundi.

name: netflix-prod > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix-prod → Namespace peru netflix-prod.

---

9. serviceaccount.yaml >> Pod identity kosam

```
apiVersion: v1
kind: ServiceAccount
metadata:
  name: netflix-sa
```

explanation:
apiVersion: v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: ServiceAccount > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada ServiceAccount object create chestunnam.

metadata: > information/details
→ ServiceAccount ki sambandhinchina basic information ikkada untundi.

name: netflix-sa > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix-sa → ServiceAccount peru netflix-sa.

---

10. role.yaml >> Permissions define cheyyadaniki

```
apiVersion: rbac.authorization.k8s.io/v1
kind: Role
metadata:
  name: netflix-role
rules:
  - apiGroups:
      - ""
    resources:
```

```
    - pods
  verbs:
    - get
    - list
```

explanation:
apiVersion: rbac.authorization.k8s.io/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: Role > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada Role object create chestunnam.

metadata: > information/details
→ Role ki sambandhinchina basic information ikkada untundi.

name: netflix-role > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix-role → Role peru netflix-role.

rules: > permissions/rules
→ Ee Role ki ye permissions undalo ikkada define chestham.

apiGroups: > API group
→ Permission ye API group resource meeda apply avvalo chepthundi.

"" > core API group
→ Empty string ante Kubernetes core API group.

resources: > ye resource?
→ Ye Kubernetes resource meeda permission undalo chepthundi.

pods > resource
→ Ikkada Pods meeda permissions istunnam.

verbs: > em actions?
→ Aa resource meeda ye actions cheyyagalado chepthundi.

get > read permission
→ Pod details ni read cheyyachu.

list > list permission
→ Pods list ni chudachu.

---

11. rolebinding.yaml >> Permissions assign cheyyadaniki

```
apiVersion: rbac.authorization.k8s.io/v1
kind: RoleBinding
metadata:
  name: netflix-rolebinding
subjects:
  - kind: ServiceAccount
    name: netflix-sa
roleRef:
  kind: Role
  name: netflix-role
  apiGroup: rbac.authorization.k8s.io
```

explanation:
apiVersion: rbac.authorization.k8s.io/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: RoleBinding > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada RoleBinding object create chestunnam.

metadata: > information/details
→ RoleBinding ki sambandhinchina basic information ikkada untundi.

name: netflix-rolebinding > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix-rolebinding → RoleBinding peru netflix-rolebinding.

subjects: > permissions evariki?
→ Role lo unna permissions evariki ivvalo chepthundi.

kind: ServiceAccount > subject type
→ Permissions receive chesedi ServiceAccount ani chepthundi.

name: netflix-sa > ServiceAccount peru
→ Permissions netflix-sa ServiceAccount ki istunnam.

roleRef: > ye Role ni assign cheyyali?
→ Ye Role lo unna permissions ni assign cheyyalo chepthundi.

kind: Role > Role type
→ Reference chestunnadi Role object ani chepthundi.

name: netflix-role > Role peru
→ netflix-role lo unna permissions ni assign chestunnam.

apiGroup: rbac.authorization.k8s.io > API group
→ Ee Role RBAC API group ki belong avtundi.

---

12. networkpolicy.yaml >> Network traffic control kosam

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
 name: netflix-backend-policy
spec:
 podSelector:
  matchLabels:
   app: backend
 policyTypes:
  - Ingress
 ingress:
  - from:
    - podSelector:
       matchLabels:
        app: frontend
   ports:
    - protocol: TCP
      port: 8080
```

explanation:
apiVersion: networking.k8s.io/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: NetworkPolicy > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada NetworkPolicy object create chestunnam.

metadata: > information/details
→ NetworkPolicy ki sambandhinchina basic information ikkada untundi.

name: netflix-backend-policy > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix-backend-policy → NetworkPolicy peru.

spec: > kavalsina setup
→ Network traffic ela allow/restrict cheyyalo ikkada cheptham.

podSelector: > ye Pods ki policy?
→ Ee policy ye Pods ki apply avvalo identify chestundi.

matchLabels: > ye label match avvali?
→ Pod labels tho match chesi correct Pods ni select chestundi.

app: backend > Pod label
→ app=backend label unna Pods ki ee policy apply avtundi.

policyTypes: > traffic type
→ Ye type network traffic ni control cheyyalo chepthundi.

Ingress > incoming traffic
→ Vere Pods nunchi backend Pods ki vachina incoming traffic ni control chestundi.

ingress: > incoming traffic rules
→ Ye incoming traffic allow cheyyalo ikkada define chestham.

from: > traffic ekkadi nunchi?
→ Traffic ye source nunchi ravachho chepthundi.

podSelector: > source Pods
→ Traffic pampinche Pods ni identify chestundi.

matchLabels: > ye label match avvali?
→ Source Pod label tho match chestundi.

app: frontend > source Pod label
→ app=frontend label unna Pods nunchi traffic allow chestunnam.

ports: > allowed ports
→ Ye port ki traffic allow cheyyalo chepthundi.

protocol: TCP > network protocol
→ TCP protocol traffic ni allow chestunnam.

port: 8080 > allowed port
→ Backend Pod port 8080 ki traffic allow chestunnam.

---

13. values.yaml >> — Helm kosam

```
replicaCount: 3
image:
  repository: mycompany/netflix
  tag: v1
service:
  type: ClusterIP
  port: 80
ingress:
  enabled: true
  host: netflix.example.com
```

explanation:
replicaCount: 3 > enni Pods?
→ Helm template lo Deployment ki 3 replicas value provide chestundi.

image: > Docker image details
→ Docker image ki sambandhinchina values ikkada untayi.

repository: mycompany/netflix > image repository
→ Docker image ye repository nunchi ravalo chepthundi.

tag: v1 > image version
→ Docker image version/tag v1 ani chepthundi.

service: > Service configuration
→ Kubernetes Service ki sambandhinchina values ikkada untayi.

type: ClusterIP > Service type
→ Service type ClusterIP ani chepthundi.

port: 80 > Service port
→ Service port 80 ani chepthundi.

ingress: > Ingress configuration
→ Ingress ki sambandhinchina values ikkada untayi.

enabled: true > Ingress enable
→ Ingress enable cheyyali ani chepthundi.

host: netflix.example.com > domain
→ Ingress use cheyyalsina domain netflix.example.com ani chepthundi.

EXTRA Yaml files

1. statefulset.yaml >>Stateful applications kosam

```
apiVersion: apps/v1
kind: StatefulSet
metadata:
 name: netflix
spec:
 serviceName: netflix-service
 replicas: 3
 selector:
  matchLabels:
   app: netflix
 template:
  metadata:
   labels:
    app: netflix
  spec:
   containers:
    - name: netflix
      image: mycompany/netflix:v1
      ports:
       - containerPort: 80
```

explanation:
apiVersion: apps/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: StatefulSet > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada StatefulSet object create chestunnam.

metadata: > information/details
→ StatefulSet ki sambandhinchina basic information ikkada untundi.

name: netflix > object peru
→ Ee object ki em peru pettalo chepthundi.
→ name: netflix → StatefulSet peru netflix.

spec: > kavalsina setup
→ StatefulSet ela work avvali ane configuration ikkada cheptham.

serviceName: netflix-service > Service peru
→ StatefulSet Pods kosam use cheyyalsina Service peru chepthundi.

replicas: 3 > enni Pods?
→ 3 Pods maintain cheyyali ani chepthundi.

selector: > ye Pods ni identify cheyyali?
→ Ee StatefulSet ye Pods ni manage cheyyalo identify chestundi.

matchLabels: > ye label match avvali?
→ Pod label tho match chesi correct Pods ni identify chestundi.

app: netflix > Pod label
→ app=netflix label unna Pods ni identify chestundi.

template: > Pod blueprint
→ Create ayye Pods ela undalo ikkada cheptham.

metadata: > Pod information/details
→ Create ayye Pod ki sambandhinchina information ikkada untundi.

labels: > Pod labels/tags
→ Pod ki labels ikkada define chestham.

app: netflix > Pod label
→ Create ayye prathi Pod ki app=netflix label vastundi.

spec: > Pod kavalsina setup
→ Pod lopala em run avvali ane configuration ikkada untundi.

containers: > Pod lo em run avvali?
→ Pod lopala run ayye container details ikkada define chestham.

name: netflix > container peru
→ Container ki netflix ane peru pettam.

image: mycompany/netflix:v1 > ye Docker image run cheyyali?
→ Container lopala mycompany/netflix:v1 Docker image run avtundi.

ports: > container ports
→ Container lo application use chese ports ikkada define chestham.

containerPort: 80 > application/container port
→ Container lo application port 80 lo run/listen chestundi ani chepthundi.

---

2. daemonset.yaml >> Prathi Node lo Pod run cheyyadaniki

```
apiVersion: apps/v1
kind: DaemonSet
metadata:
 name: netflix-agent
spec:
 selector:
  matchLabels:
   app: netflix-agent
 template:
  metadata:
   labels:
    app: netflix-agent
  spec:
   containers:
    - name: netflix-agent
      image: mycompany/netflix-agent:v1
```

explanation:
apiVersion: apps/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: DaemonSet > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada DaemonSet object create chestunnam.

metadata: > information/details
→ DaemonSet ki sambandhinchina basic information ikkada untundi.

name: netflix-agent > object peru
→ Ee object ki netflix-agent ane peru pettam.

spec: > kavalsina setup
→ DaemonSet ela work avvali ane configuration ikkada cheptham.

selector: > ye Pods ni identify cheyyali?
→ Ee DaemonSet ye Pods ni manage cheyyalo identify chestundi.

matchLabels: > ye label match avvali?
→ Pod label tho match chesi correct Pods ni identify chestundi.

app: netflix-agent > Pod label
→ app=netflix-agent label unna Pods ni identify chestundi.

template: > Pod blueprint
→ Kubernetes create chese Pod ela undalo ikkada cheptham.

metadata: > Pod information/details
→ Create ayye Pod ki sambandhinchina information ikkada untundi.

labels: > Pod labels/tags
→ Pod ki labels ikkada define chestham.

app: netflix-agent > Pod label
→ Create ayye Pod ki app=netflix-agent label vastundi.

spec: > Pod kavalsina setup
→ Pod lopala em run avvali ane configuration ikkada untundi.

containers: > Pod lo em run avvali?
→ Pod lopala run ayye container details ikkada define chestham.

name: netflix-agent > container peru
→ Container ki netflix-agent ane peru pettam.

image: mycompany/netflix-agent:v1 > ye Docker image run cheyyali?
→ Container lopala mycompany/netflix-agent:v1 Docker image run avtundi.

---

3. job.yaml >> One-time task run cheyyadaniki

```
apiVersion: batch/v1
kind: Job
metadata:
  name: netflix-job
spec:
  template:
    spec:
      containers:
        - name: netflix-job
          image: mycompany/netflix-job:v1
      restartPolicy: Never
```

explanation:
apiVersion: batch/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: Job > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada Job object create chestunnam.

metadata: > information/details
→ Job ki sambandhinchina basic information ikkada untundi.

name: netflix-job > object peru
→ Ee object ki netflix-job ane peru pettam.

spec: > kavalsina setup
→ Job ela run avvali ane configuration ikkada cheptham.

template: > Pod blueprint
→ Job create chese Pod ela undalo ikkada cheptham.

spec: > Pod kavalsina setup
→ Pod lopala em run avvali ane configuration ikkada untundi.

containers: > Pod lo em run avvali?
→ Pod lopala run ayye container details ikkada define chestham.

name: netflix-job > container peru
→ Container ki netflix-job ane peru pettam.

image: mycompany/netflix-job:v1 > ye Docker image run cheyyali?
→ Container lopala mycompany/netflix-job:v1 Docker image run avtundi.

restartPolicy: Never > restart rule
→ Container complete ayyaka fail ayina appudu Pod ni automatically restart cheyyakudadhu ani chepthundi.

---

4. cronjob.yaml >> Scheduled tasks run cheyyadaniki

```
apiVersion: batch/v1
kind: CronJob
metadata:
  name: netflix-backup
spec:
  schedule: "0 2 * * *"
  jobTemplate:
    spec:
      template:
        spec:
          containers:
            - name: netflix-backup
              image: mycompany/netflix-backup:v1
          restartPolicy: Never
```

explanation:
apiVersion: batch/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: CronJob > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada CronJob object create chestunnam.

metadata: > information/details
→ CronJob ki sambandhinchina basic information ikkada untundi.

name: netflix-backup > object peru
→ Ee object ki netflix-backup ane peru pettam.

spec: > kavalsina setup
→ CronJob eppudu, ela run avvali ane configuration ikkada cheptham.

schedule: "0 2 * * *" > schedule/time
→ Job daily 2:00 AM ki run avvali ani chepthundi.

jobTemplate: > Job blueprint
→ Schedule time vachinappudu create ayye Job ela undalo ikkada cheptham.

spec: > Job kavalsina setup
→ Job configuration ikkada untundi.

template: > Pod blueprint
→ Job create chese Pod ela undalo ikkada cheptham.

spec: > Pod kavalsina setup
→ Pod lopala em run avvali ane configuration ikkada untundi.

containers: > Pod lo em run avvali?
→ Pod lopala run ayye container details ikkada define chestham.

name: netflix-backup > container peru
→ Container ki netflix-backup ane peru pettam.

image: mycompany/netflix-backup:v1 > ye Docker image run cheyyali?
→ Container lopala backup Docker image run avtundi.

restartPolicy: Never > restart rule
→ Job complete ayyaka Pod ni automatically restart cheyyakudadhu ani chepthundi.

---

5. pv.yaml >> Persistent storage provide cheyyadaniki

```
apiVersion: v1
kind: PersistentVolume
metadata:
  name: netflix-pv
spec:
  capacity:
    storage: 10Gi
  accessModes:
    - ReadWriteOnce
  hostPath:
    path: /data/Netflix
```

explanation:
apiVersion: v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: PersistentVolume > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada PersistentVolume object create chestunnam.

metadata: > information/details
→ PersistentVolume ki sambandhinchina basic information ikkada untundi.

name: netflix-pv > object peru
→ Ee object ki netflix-pv ane peru pettam.

spec: > kavalsina setup
→ Storage ela undalo ane configuration ikkada cheptham.

capacity: > storage capacity
→ Ee volume lo entha storage available undo chepthundi.

storage: 10Gi > storage size
→ Ee PersistentVolume capacity 10Gi ani chepthundi.

accessModes: > storage access mode
→ Storage ni ela access cheyyalo chepthundi.

ReadWriteOnce > access mode
→ Oka node nunchi volume ni read/write cheyyadaniki allow chestundi.

hostPath: > storage location
→ Node lo storage ekkada undalo chepthundi.

path: /data/netflix > storage path
→ Node lo /data/netflix location ni storage kosam use chestundi.

---

6. storageclass.yaml >> Storage provisioning rules kosam

```
apiVersion: storage.k8s.io/v1
kind: StorageClass
metadata:
  name: netflix-storage
provisioner: kubernetes.io/no-provisioner
volumeBindingMode: WaitForFirstConsumer
```

explanation:
apiVersion: storage.k8s.io/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: StorageClass > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada StorageClass object create chestunnam.

metadata: > information/details
→ StorageClass ki sambandhinchina basic information ikkada untundi.

name: netflix-storage > object peru
→ Ee object ki netflix-storage ane peru pettam.

provisioner: kubernetes.io/no-provisioner > storage provisioner
→ Storage ni evaru provision cheyyalo chepthundi.
→ no-provisioner ante automatic storage create cheyyadu.

volumeBindingMode: WaitForFirstConsumer > volume binding rule
→ Pod storage ni actually use cheyyadaniki vachina tarvatha volume binding jaragali ani chepthundi.

---

7. poddisruptionbudget.yaml >> Minimum Pods available ga maintain cheyyadaniki

```
apiVersion: policy/v1
kind: PodDisruptionBudget
metadata:
  name: netflix-pdb
spec:
  minAvailable: 2
  selector:
    matchLabels:
      app: Netflix
```

explanation:
apiVersion: policy/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: PodDisruptionBudget > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada PodDisruptionBudget object create chestunnam.

metadata: > information/details
→ PDB ki sambandhinchina basic information ikkada untundi.

name: netflix-pdb > object peru
→ Ee object ki netflix-pdb ane peru pettam.

spec: > kavalsina setup
→ Enni Pods available ga undalo ikkada cheptham.

minAvailable: 2 > minimum available Pods
→ Voluntary disruption time lo minimum 2 Pods available ga undali ani chepthundi.

selector: > ye Pods ki policy?
→ Ee PDB ye Pods ki apply avvalo identify chestundi.

matchLabels: > ye label match avvali?
→ Pod labels tho match chesi correct Pods ni select chestundi.

app: netflix > Pod label
→ app=netflix label unna Pods ki ee PDB apply avtundi.

---

8. resourcequota.yaml >>Namespace resource limits kosam

```
apiVersion: v1
kind: ResourceQuota
metadata:
 name: netflix-quota
spec:
 hard:
   pods: "10"
   requests.cpu: "4"
   requests.memory: 8Gi
```

explanation:
apiVersion: v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: ResourceQuota > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada ResourceQuota object create chestunnam.

metadata: > information/details
→ ResourceQuota ki sambandhinchina basic information ikkada untundi.

name: netflix-quota > object peru
→ Ee object ki netflix-quota ane peru pettam.

spec: > kavalsina setup
→ Namespace lo resources ki entha limit undalo ikkada cheptham.

hard: > resource limits
→ Maximum ga entha resource use cheyyavachho ikkada define chestham.

pods: "10" > maximum Pods
→ Ee Namespace lo maximum 10 Pods create cheyyavachu.

requests.cpu: "4" > CPU limit
→ Total CPU requests maximum 4 CPU units varaku allow chestundi.

requests.memory: 8Gi > memory limit
→ Total memory requests maximum 8Gi varaku allow chestundi.

9. limitrange.yaml >> Container resource limits/defaults kosam

```
apiVersion: v1
kind: LimitRange
metadata:
  name: netflix-limits
spec:
  limits:
    - type: Container
      default:
        cpu: "500m"
        memory: "512Mi"
      defaultRequest:
        cpu: "250m"
        memory: "256Mi"
```

explanation:

apiVersion: v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: LimitRange > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada LimitRange object create chestunnam.

metadata: > information/details
→ LimitRange ki sambandhinchina basic information ikkada untundi.

name: netflix-limits > object peru
→ Ee object ki netflix-limits ane peru pettam.

spec: > kavalsina setup
→ Containers ki default resource limits/requests ela undalo ikkada cheptham.

limits: > resource limits/settings
→ Container resource settings ikkada define chestham.

type: Container > ye resource ki?
→ Ee limits Containers ki apply avtayi.

default: > default limit
→ Container resource limit specify cheyyakapothe ee values use avtayi.

cpu: "500m" > CPU limit
→ Container default CPU limit 500m.

memory: "512Mi" > memory limit
→ Container default memory limit 512Mi.

defaultRequest: > default resource request
→ Container resource request specify cheyyakapothe ee values use avtayi.

cpu: "250m" > CPU request
→ Container default CPU request 250m.

memory: "256Mi" > memory request
→ Container default memory request 256Mi.

---

10. clusterrole.yaml >> Cluster-level permissions define cheyyadaniki

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRole
metadata:
  name: netflix-cluster-role
rules:
  - apiGroups:
      - ""
    resources:
      - pods
    verbs:
      - get
      - list
```

explanation:

apiVersion: rbac.authorization.k8s.io/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: ClusterRole > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada ClusterRole object create chestunnam.

metadata: > information/details
→ ClusterRole ki sambandhinchina basic information ikkada untundi.

name: netflix-cluster-role > object peru
→ Ee object ki netflix-cluster-role ane peru pettam.

rules: > permissions/rules
→ ClusterRole ki ye permissions undalo ikkada define chestham.

apiGroups: > API group
→ Permission ye API group resource meeda apply avvalo chepthundi.

"" > core API group
→ Empty string ante Kubernetes core API group.

resources: > ye resource?
→ Ye Kubernetes resource meeda permission undalo chepthundi.

pods > resource
→ Ikkada Pods meeda permissions istunnam.

verbs: > em actions?
→ Aa resource meeda ye actions cheyyagalado chepthundi.

get > read permission
→ Pod details ni read cheyyachu.

list > list permission
→ Pods list ni chudachu.

---

11. clusterrolebinding.yaml >> Cluster-level permissions assign cheyyadaniki

```
apiVersion: rbac.authorization.k8s.io/v1
kind: ClusterRoleBinding
metadata:
  name: netflix-cluster-rolebinding
subjects:
  - kind: ServiceAccount
    name: netflix-sa
    namespace: netflix-prod
roleRef:
  kind: ClusterRole
  name: netflix-cluster-role
  apiGroup: rbac.authorization.k8s.io
```

explanation:

apiVersion: rbac.authorization.k8s.io/v1 > version/rules
→ Kubernetes ki ee object ni ye API version/rules tho handle cheyyalo chepthundi.

kind: ClusterRoleBinding > em object?
→ Kubernetes lo ye type object create cheyyalo chepthundi.
→ Ikkada ClusterRoleBinding object create chestunnam.

metadata: > information/details
→ ClusterRoleBinding ki sambandhinchina basic information ikkada untundi.

name: netflix-cluster-rolebinding > object peru
→ Ee object ki netflix-cluster-rolebinding ane peru pettam.

subjects: > permissions evariki?
→ ClusterRole lo unna permissions evariki ivvalo chepthundi.

kind: ServiceAccount > subject type
→ Permissions receive chesedi ServiceAccount ani chepthundi.

name: netflix-sa > ServiceAccount peru
→ Permissions netflix-sa ServiceAccount ki istunnam.

namespace: netflix-prod > ServiceAccount namespace
→ netflix-sa ServiceAccount netflix-prod Namespace lo undi ani chepthundi.

roleRef: > ye Role ni assign cheyyali?
→ Ye ClusterRole permissions ni assign cheyyalo chepthundi.

kind: ClusterRole > Role type
→ Reference chestunnadi ClusterRole object ani chepthundi.

name: netflix-cluster-role > ClusterRole peru
→ netflix-cluster-role lo unna permissions ni assign chestunnam.

apiGroup: rbac.authorization.k8s.io > API group
→ Ee ClusterRole RBAC API group ki belong avtundi.